# File Guide

LinkedIn hiring loop — reverse-engineered

DOTE 6751 Database Systems · Group Project · November 2026

The map of the package: every file, how to rebuild, how to verify, and what to hand over.

## 1. The map

*Generated by scanning the folder, so the list and the files agree.*

```
LinkedIn_交付/
  File_Guide.docx   (42 KB)
  File_Guide.pdf   (201 KB)
  File_Guide_CN.docx   (43 KB)
  File_Guide_CN.pdf   (396 KB)
  README.md   (1 KB)
  build/
    README.md   (2 KB)
    build_all.py   (2 KB)
    build_deck.py   (44 KB)
    build_file_guide.py   (23 KB)
    build_narrative_docx.py   (15 KB)
    build_queries.py   (4 KB)
    erd.py   (15 KB)
    fault_injection.py   (5 KB)
    gen_normalization_xlsx.py   (24 KB)
    gen_seed.py   (42 KB)
    queries_template.sql   (23 KB)
    run_queries.py   (2 KB)
    verify_data.py   (8 KB)
  design/
    Constraint_Inventory.md   (10 KB)
    Design_Rationale.md   (12 KB)
    Entity_Relationship_Catalog.md   (13 KB)
    Normalization_Proof.xlsx   (29 KB)
    Product_Spec_CN.docx   (42 KB)
    Product_Spec_CN.md   (8 KB)
    Product_Spec_EN.docx   (41 KB)
    Product_Spec_EN.md   (8 KB)
    Relational_Schema.md   (11 KB)
    Requirements.md   (8 KB)
  erd/
    ERD_conceptual.png   (359 KB)
    ERD_conceptual.svg   (83 KB)
    ERD_logical_full.png   (267 KB)
    ERD_logical_full.svg   (179 KB)
    ERD_module_demand.png   (135 KB)
    ERD_module_demand.svg   (92 KB)
    ERD_module_hiring_loop.png   (152 KB)
    ERD_module_hiring_loop.svg   (103 KB)
    ERD_module_identity.png   (124 KB)
    ERD_module_identity.svg   (97 KB)
  query_results/
    _consistency_report.json   (3 KB)
    _summary.json   (6 KB)
    q01_result.csv   (<1 KB)
```

```
  q01_result.png   (84 KB)
  q02_result.csv   (<1 KB)
  q02_result.png   (100 KB)
  q03_result.csv   (2 KB)
  q04_result.csv   (<1 KB)
  q05_result.csv   (<1 KB)
  q05_result.png   (70 KB)
  q06_result.csv   (<1 KB)
  q07_result.csv   (<1 KB)
  q08_result.csv   (<1 KB)
  q09_result.csv   (<1 KB)
  q09_result.png   (76 KB)
  q10_result.csv   (<1 KB)
  q11_result.csv   (<1 KB)
  q12_result.csv   (<1 KB)
  q12_result.png   (73 KB)
  q13_result.csv   (<1 KB)
  q14_result.csv   (<1 KB)
  q15_result.csv   (1 KB)
  q16_result.csv   (<1 KB)
  q17_result.csv   (1 KB)
  q18_result.csv   (<1 KB)
  q18_result.png   (75 KB)
slides/
  LinkedIn.pptx   (1001 KB)
  LinkedIn_CN.pptx   (1003 KB)
  LinkedIn_Narrative.docx   (41 KB)
  LinkedIn_Narrative.pdf   (95 KB)
  Presentation_Script.md   (10 KB)
  QA_Prep.md   (8 KB)
  Task_Allocation.md   (7 KB)
sql/
  00_run_all.sql   (<1 KB)
  01_schema.sql   (17 KB)
  02_seed.sql   (292 KB)
  03_queries.sql   (23 KB)
  04_views_triggers.sql   (10 KB)
  05_transactions.sql   (4 KB)
  06_checks.sql   (5 KB)
  linkedin_db_full.sql   (273 KB)
```

## 2. Every file, and what it is for

Grouped by folder. What matters is when you would open it.

### Root

| File | Size | What it is | What is inside |
|---|---|---|---|
| File_Guide.docx | 42 KB | — | — |
| File_Guide.pdf | 201 KB | — | — |
| File_Guide_CN.docx | 43 KB | — | — |
| File_Guide_CN.pdf | 396 KB | — | — |
| README.md | 1 KB | Quick orientation | Three sentences on what the package is and where to start. |

## build/

| File | Size | What it is | What is inside |
|---|---|---|---|
| README.md | 2 KB | Quick orientation | Three sentences on what the package is and where to start. |
| build_all.py | 2 KB | Build script | Regenerates the artefact; not a deliverable. |
| build_deck.py | 44 KB | Build script | Regenerates the artefact; not a deliverable. |
| build_file_guide.py | 23 KB | Build script | Regenerates the artefact; not a deliverable. |
| build_narrative_docx.py | 15 KB | Build script | Regenerates the artefact; not a deliverable. |
| build_queries.py | 4 KB | Build script | Regenerates the artefact; not a deliverable. |
| erd.py | 15 KB | Build script | Regenerates the artefact; not a deliverable. |
| fault_injection.py | 5 KB | Build script | Regenerates the artefact; not a deliverable. |
| gen_normalization_xlsx.py | 24 KB | Build script | Regenerates the artefact; not a deliverable. |
| gen_seed.py | 42 KB | Build script | Regenerates the artefact; not a deliverable. |
| queries_template.sql | 23 KB | Build script | Regenerates the artefact; not a deliverable. |
| run_queries.py | 2 KB | Build script | Regenerates the artefact; not a deliverable. |
| verify_data.py | 8 KB | Build script | Regenerates the artefact; not a deliverable. |

## design/

| File | Size | What it is | What is inside |
|---|---|---|---|
| Constraint_Inventory.md | 10 KB | Constraint inventory | Every rule, its type, where it is enforced, and what happens when it breaks. |
| Design_Rationale.md | 12 KB | Design rationale | Why these tables, why the weak entity, why derived values are stored, what was rejected. |
| Entity_Relationship_Catalog.md | 13 KB | E-R catalogue | Every entity and relationship written out: cardinality, participation, attributes. |
| Normalization_Proof.xlsx | 29 KB | Normalization proof | Functional dependencies, candidate keys and the 3NF/BCNF argument, per table. |
| Product_Spec_CN.docx | 42 KB | Product spec (CN) | Positioning, vision, substitutes, scenarios, features, economics, metrics, risks, roadmap. |
| Product_Spec_CN.md | 8 KB | Product spec source (CN) | The same document as Markdown. |
| Product_Spec_EN.docx | 41 KB | Product spec (EN) | The same document in English. |
| Product_Spec_EN.md | 8 KB | Product spec source (EN) | The same document as Markdown. |
| Relational_Schema.md | 11 KB | Relational schema | The E-R design translated into relations, with keys and mapping rules. |
| Requirements.md | 8 KB | Requirements and scope | The problem, the actors, the functional requirements, scope and assumptions. |

## erd/

| File | Size | What it is | What is inside |
|---|---|---|---|
| ERD_conceptual.png | 359 KB | Conceptual E-R | Entities, relationship diamonds and cardinality. |
| ERD_conceptual.svg | 83 KB | Vector version | Scales without blur; opens in a browser. |
| ERD_logical_full.png | 267 KB | Logical schema | Every table with columns, keys and connectors. |
| ERD_logical_full.svg | 179 KB | Vector version | Scales without blur; opens in a browser. |
| ERD_module_demand.png | 135 KB | Module E-R | A zoomed view of that module. |
| ERD_module_demand.svg | 92 KB | Vector version | Scales without blur; opens in a browser. |
| ERD_module_hiring_loop.png | 152 KB | Module E-R | A zoomed view of that module. |
| ERD_module_hiring_loop.svg | 103 KB | Vector version | Scales without blur; opens in a browser. |
| ERD_module_identity.png | 124 KB | Module E-R | A zoomed view of that module. |
| ERD_module_identity.svg | 97 KB | Vector version | Scales without blur; opens in a browser. |

## query_results/

| File | Size | What it is | What is inside |
|---|---|---|---|
| _consistency_report.json | 3 KB | Consistency report | Results of the cross-table invariants. |
| _summary.json | 6 KB | Query summary | Rows, timing and columns per query. |
| q01_result.csv | <1 KB | Query 1 result | The full output of that query on the seeded instance. |
| q01_result.png | 84 KB | Query 1 result image | The result rendered as a table image. |
| q02_result.csv | <1 KB | Query 2 result | The full output of that query on the seeded instance. |
| q02_result.png | 100 KB | Query 2 result image | The result rendered as a table image. |
| q03_result.csv | 2 KB | Query 3 result | The full output of that query on the seeded instance. |
| q04_result.csv | <1 KB | Query 4 result | The full output of that query on the seeded instance. |
| q05_result.csv | <1 KB | Query 5 result | The full output of that query on the seeded instance. |
| q05_result.png | 70 KB | Query 5 result image | The result rendered as a table image. |
| q06_result.csv | <1 KB | Query 6 result | The full output of that query on the seeded instance. |
| q07_result.csv | <1 KB | Query 7 result | The full output of that query on the seeded instance. |
| q08_result.csv | <1 KB | Query 8 result | The full output of that query on the seeded instance. |
| q09_result.csv | <1 KB | Query 9 result | The full output of that query on the seeded instance. |
| q09_result.png | 76 KB | Query 9 result image | The result rendered as a table image. |
| q10_result.csv | <1 KB | Query 10 result | The full output of that query on the seeded instance. |
| q11_result.csv | <1 KB | Query 11 result | The full output of that query on the seeded instance. |
| q12_result.csv | <1 KB | Query 12 result | The full output of that query on the seeded instance. |
| q12_result.png | 73 KB | Query 12 result image | The result rendered as a table image. |
| q13_result.csv | <1 KB | Query 13 result | The full output of that query on the seeded instance. |
| q14_result.csv | <1 KB | Query 14 result | The full output of that query on the seeded instance. |

| | | | |
|---|---|---|---|
| q15_result.csv | 1 KB | Query 15 result | The full output of that query on the seeded instance. |
| q16_result.csv | <1 KB | Query 16 result | The full output of that query on the seeded instance. |
| q17_result.csv | 1 KB | Query 17 result | The full output of that query on the seeded instance. |
| q18_result.csv | <1 KB | Query 18 result | The full output of that query on the seeded instance. |
| q18_result.png | 75 KB | Query 18 result image | The result rendered as a table image. |

### slides/

| File | Size | What it is | What is inside |
|---|---|---|---|
| LinkedIn.pptx | 1001 KB | Deck (EN) | Fourteen slides with speaker notes. |
| LinkedIn_CN.pptx | 1003 KB | Deck (CN) | The same deck in Chinese. |
| LinkedIn_Narrative.docx | 41 KB | Narrative | About 1,600 words across six sections. |
| LinkedIn_Narrative.pdf | 95 KB | Narrative PDF | The same text. |
| Presentation_Script.md | 10 KB | Speaking script | What to say on each slide, with timings and speaker split. |
| QA_Prep.md | 8 KB | Q&A preparation | Likely questions with answers short enough to say aloud. |
| Task_Allocation.md | 7 KB | Task allocation | Seven technical tasks with difficulty and hours; no presentation role. |

### sql/

| File | Size | What it is | What is inside |
|---|---|---|---|
| 00_run_all.sql | <1 KB | Rebuild script | Calls the other scripts in the only correct order. |
| 01_schema.sql | 17 KB | DDL | Creates the database and every table with keys, checks and indexes. |
| 02_seed.sql | 292 KB | Seed data | Deterministic data; ends with a reconciliation statement. |
| 03_queries.sql | 23 KB | Eighteen queries | Three families; each statement runs on its own. |
| 04_views_triggers.sql | 10 KB | Views and triggers | Read models and the triggers that enforce business rules. |
| 05_transactions.sql | 4 KB | Transaction episodes | Self-contained episodes, rolled back, so the data is unchanged. |
| 06_checks.sql | 5 KB | Acceptance checks | Twelve checks; every row must read PASS. |
| linkedin_db_full.sql | 273 KB | Full dump | Schema, data, views and triggers in one file. |

## 3. How to rebuild the database

Navicat: Run SQL File → sql/linkedin_db_full.sql.

Or in order: 01_schema → 02_seed → 04_views_triggers → 06_checks → 03_queries.

Command line: mysql -u root -p < 00_run_all.sql

Order matters: the seed contains applications withdrawn after interviews, which the guards would refuse.

## 4. How to verify

Eight steps; the first six take three minutes.

| # | Check | How | A pass looks like |
|---|---|---|---|
| 1 | Service running | MySQL80 Running | Started |
| 2 | Client connects | Test Connection | Success |
| 3 | Schema complete | Restore and expand | 12 tables, 4 views, 4 triggers |
| 4 | Data arrived | Count members / applications / interviews | 300 / 250 / 210 |
| 5 | Acceptance script | Run 06_checks.sql | 12 rows, all PASS |
| 6 | Queries answer | Run 03_queries.sql | q01 seven rows, q09 twelve |
| 7 | Constraints refuse | Run the three below | Each fails |
| 8 | Invariants hold | Open _consistency_report.json | 22 of 22 pass |

These three are meant to fail. If one succeeds, a constraint is missing.

**Referral from outside**

```
INSERT INTO referral (application_id, referrer_id, relationship, referred_at, bonus_amount,
bonus_status)
VALUES (1, 1, 'OTHER', NOW(), 0, 'NONE');
```

Expected: error 1644, referrer not employed

**Interview too early**

```
INSERT INTO interview (application_id, round_no, interview_type, scheduled_at, duration_min,
interviewer_id, outcome)
VALUES (9, 1, 'PHONE', NOW() + INTERVAL 2 DAY, 30, 1, 'PENDING');
```

Expected: error 1644, stage not reached

**Duplicate application**

```
INSERT INTO application (job_id, member_id, source_channel, status, applied_at)
VALUES (1, 1, 'DIRECT', 'SUBMITTED', NOW());
```

Expected: error 1062, duplicate key

## 5. What was checked before delivery

Twelve checks, all executed.

| # | Check | Violations | Result |
|---|---|---|---|
| C01 | Twelve tables present | 12 | PASS |
| C02 | Four views and four triggers | 8 | PASS |
| C03 | No decision before application | 0 | PASS |
| C04 | Terminal rows have a date | 0 | PASS |

| | | | |
|---|---|---|---|
| C05 | In-flight rows have none | 0 | PASS |
| C06 | Hires within openings | 0 | PASS |
| C07 | No self-connection | 0 | PASS |
| C08 | No duplicate connections | 0 | PASS |
| C09 | Proficiency in 1–5 | 0 | PASS |
| C10 | Scores in 1–5 | 0 | PASS |
| C11 | Referrals from inside | 0 | PASS |
| C12 | One current employer each | 0 | PASS |

## 6. Decisions worth knowing

**Twelve tables**

Eleven entity sets plus the self-referential connection; the current employer is not a column.

**The current employer is derived**

experience.is_current plus a unique index on a generated column.

**Referral is 1:1 with its application**

It stores only the referrer and the bonus.

**Guards live in the database**

Referrer employment and interview stage cannot be bypassed by any client.

**The data is synthetic**

Plausible distributions, not empirical; every finding describes the seeded instance.